



NATIONAL UNIVERSITY
of Computer & Emerging Sciences

Subject: Digital Signal Processing

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Example 2.3.3

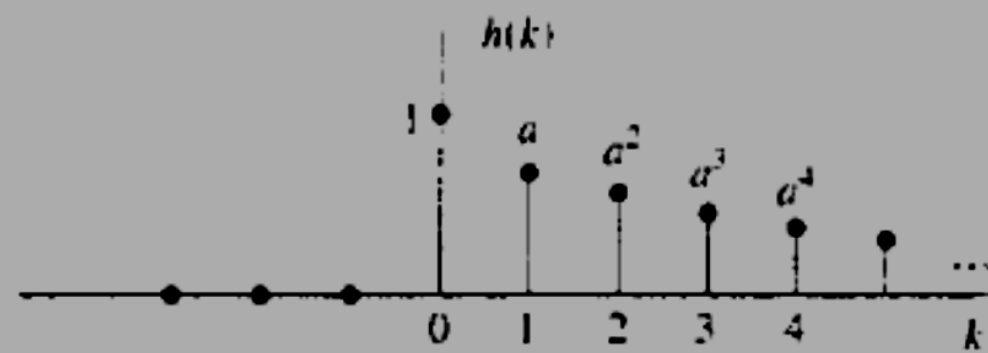
Determine the output $y(n)$ of a relaxed linear time-invariant system with impulse response

$$h(n) = a^n u(n), |a| < 1$$

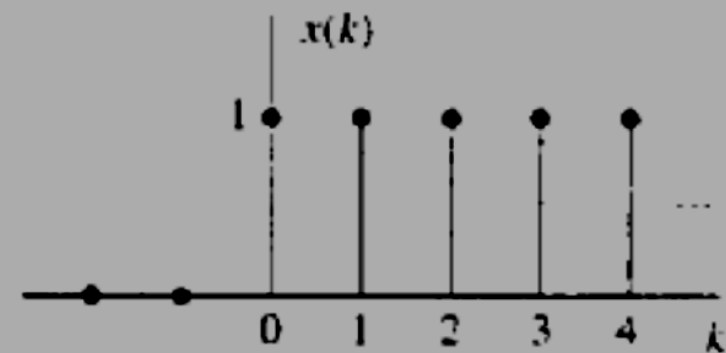
when the input is a unit step sequence, that is,

$$x(n) = u(n)$$

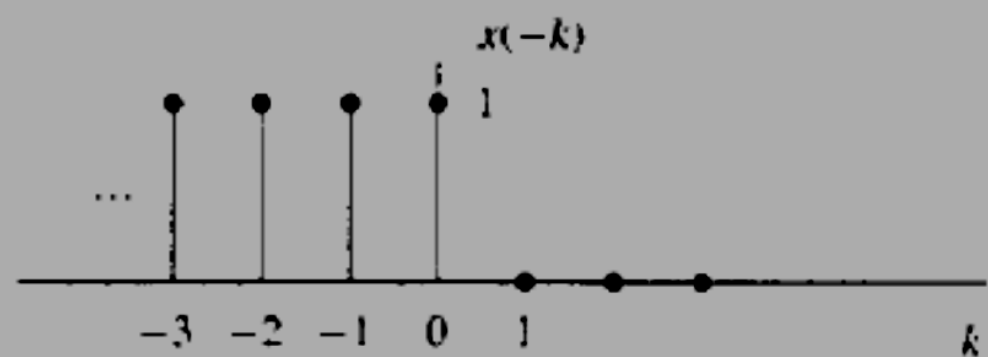
Solution In this case both $h(n)$ and $x(n)$ are infinite-duration sequences. We use the form of the convolution formula given by (2.3.28) in which $x(k)$ is folded. The



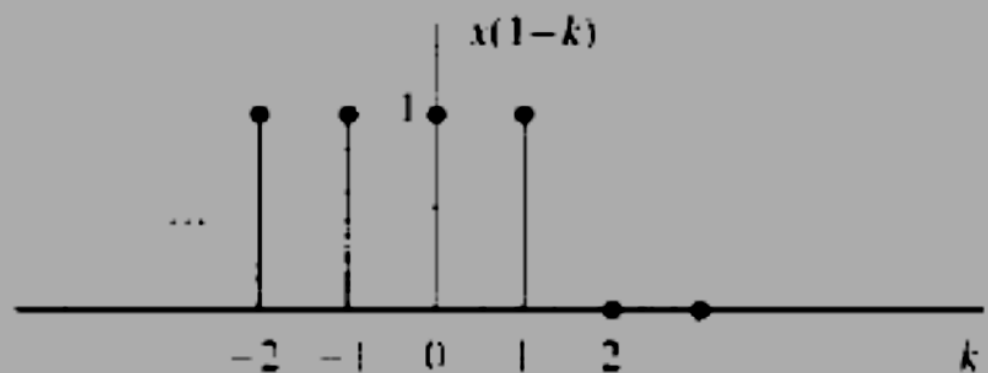
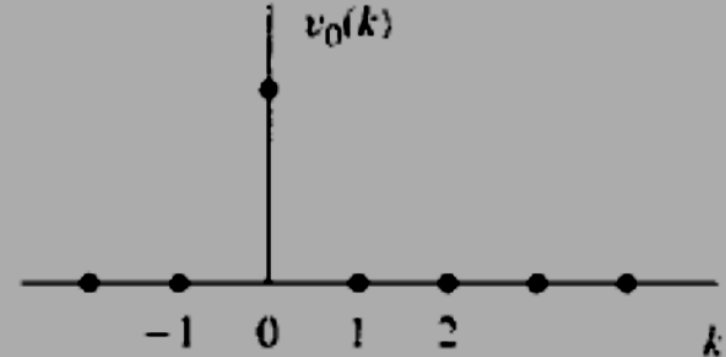
(a)



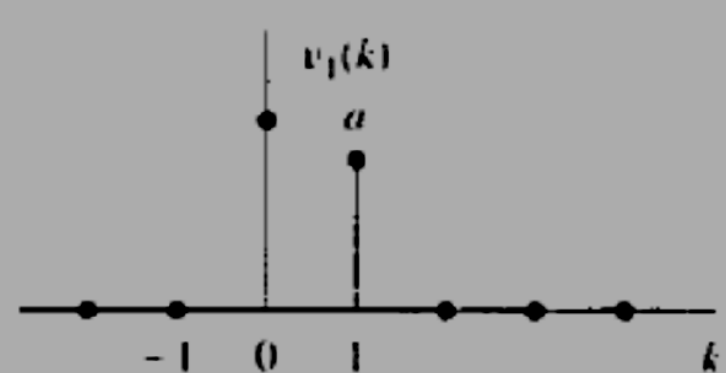
(b)



(c)



(d)



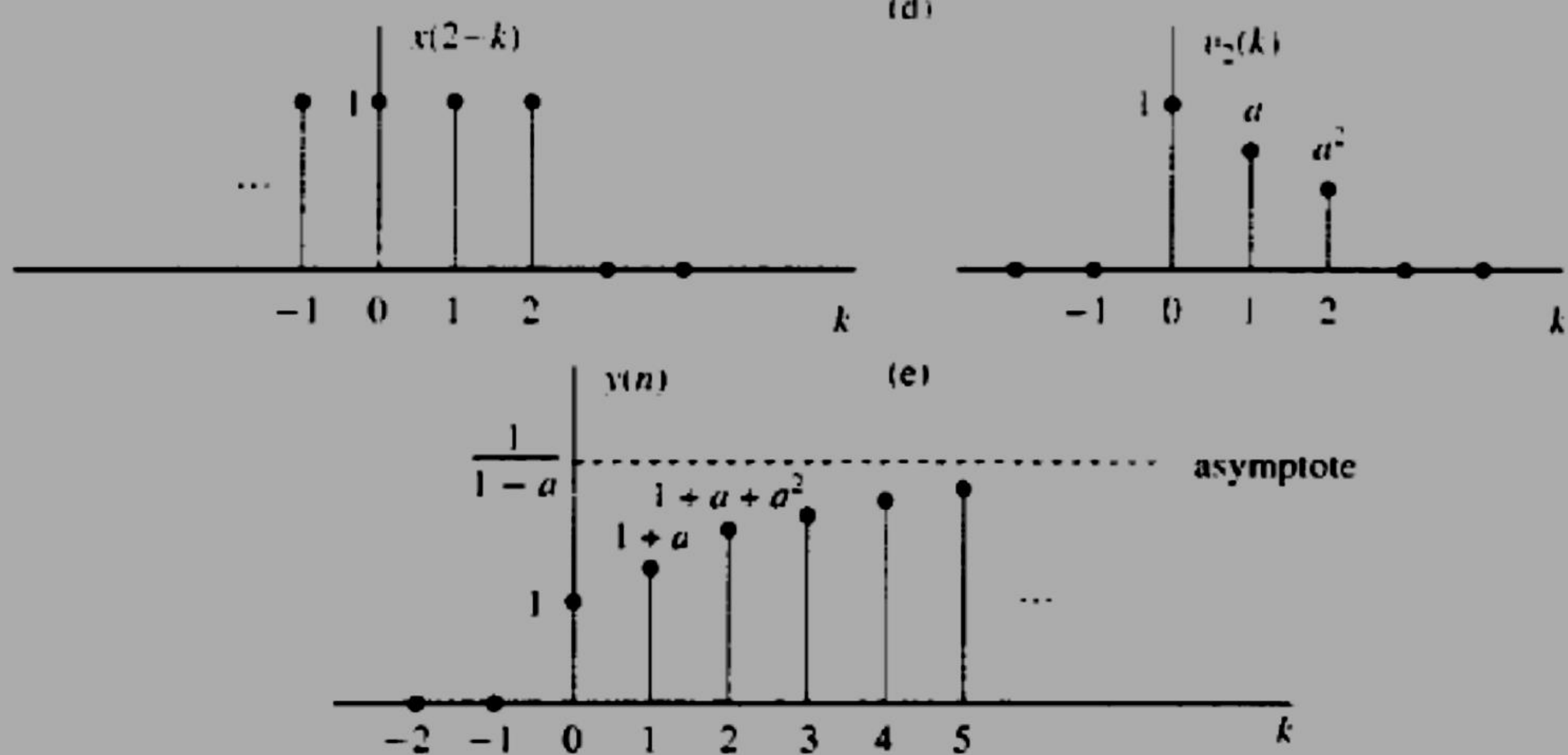


Figure 2.24 Graphical computation of convolution in Example 2.3.3.

sequences $h(k)$, $x(k)$, and $x(-k)$ are shown in Fig. 2.24. The product sequences $v_0(k)$, $v_1(k)$, and $v_2(k)$ corresponding to $x(-k)h(k)$, $x(1-k)h(k)$, and $x(2-k)h(k)$ are illustrated in Fig. 2.24c, d, and e, respectively. Thus we obtain the outputs

$$y(0) = 1$$

$$y(1) = 1 + a$$

$$y(2) = 1 + a + a^2$$

Clearly, for $n > 0$, the output is

$$\begin{aligned} y(n) &= 1 + a + a^2 + \cdots + a^n \\ &= \frac{1 - a^{n+1}}{1 - a} \end{aligned} \quad (2.3.29)$$

On the other hand, for $n < 0$, the product sequences consist of all zeros. Hence

$$y(n) = 0 \quad n < 0$$

A graph of the output $y(n)$ is illustrated in Fig. 2.24f for the case $0 < a < 1$. Note the exponential rise in the output as a function of n . Since $|a| < 1$, the final value of the output as n approaches infinity is

$$y(\infty) = \lim_{n \rightarrow \infty} y(n) = \frac{1}{1 - a} \quad (2.3.30)$$

To summarize, the convolution formula provides us with a means for computing the response of a relaxed, linear time-invariant system to any arbitrary input signal $x(n)$. It takes one of two equivalent forms, either (2.3.17) or (2.3.28), where $x(n)$ is the input signal to the system, $h(n)$ is the impulse response of the system, and $y(n)$ is the *output* of the system in response to the input signal $x(n)$. The evaluation of the convolution formula involves four operations, namely: *folding* either the impulse response as specified by (2.3.17) or the input sequence as specified by (2.3.28) to yield either $h(-k)$ or $x(-k)$, respectively, *shifting* the folded sequence by n units in time to yield either $h(n - k)$ or $x(n - k)$, *multiplying* the two sequences to yield the product sequence, either $x(k)h(n - k)$ or $x(n - k)h(k)$, and finally *summing* all the values in the product sequence to yield the output $y(n)$ of the system at time n . The folding operation is done only once. However, the other three operations are repeated for all possible shifts $-\infty < n < \infty$ in order to obtain $y(n)$ for $-\infty < n < \infty$.

Example 2.3.4

Determine the impulse response for the cascade of two linear time-invariant systems having impulse responses

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n)$$

and

$$h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

Solution To determine the overall impulse response of the two systems in cascade, we simply convolve $h_1(n)$ with $h_2(n)$. Hence

$$h(n) = \sum_{k=-\infty}^{\infty} h_1(k)h_2(n-k)$$

where $h_2(n)$ is folded and shifted. We define the product sequence

$$\begin{aligned} v_n(k) &= h_1(k)h_2(n-k) \\ &= \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} \end{aligned}$$

which is nonzero for $k \geq 0$ and $n-k \geq 0$ or $n \geq k \geq 0$. On the other hand, for $n < 0$, we have $v_n(k) = 0$ for all k , and hence

$$h(n) = 0, n < 0$$

For $n \geq k \geq 0$, the sum of the values of the product sequence $v_n(k)$ over all k yields

$$\begin{aligned}h(n) &= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k} \\&= \left(\frac{1}{4}\right)^n \sum_{k=0}^n 2^k \\&= \left(\frac{1}{4}\right)^n (2^{n+1} - 1) \\&= \left(\frac{1}{2}\right)^n [2 - \left(\frac{1}{2}\right)^n], n \geq 0\end{aligned}$$

$$h(n) = \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{4}\right)^{n-k}$$

$$\left(\frac{1}{4}\right)^{n-k} = \left(\frac{1}{2^2}\right)^{n-k} = \left(\frac{1}{2}\right)^{2(n-k)}$$

$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2(n-k)}$$

$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{2n-2k}$$

Since $\left(\frac{1}{2}\right)^{2n}$ is independent of $\sum_k$,
factor it out:

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^n \left(\frac{1}{2}\right)^k \cdot \left(\frac{1}{2}\right)^{-2k} = \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^n \left(\frac{1}{2}\right)^{-k}$$

$$= \left(\frac{1}{2}\right)^{2n} \sum_{k=0}^n \left(\frac{1}{2}\right)^{-k}$$

$$= \sum_{k=0}^n 2^k = \frac{1-2^{n+1}}{1-2} = 2^{n+1} - 1$$

By sum formula for geom-
etric series.

$$= \left(\frac{1}{2}\right)^{2n} (2^{n+1} - 1) = \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n (2^{n+1} - 1)$$

$$= \left(\frac{1}{2}\right)^n \left(\frac{1}{2}\right)^n [2^n \cdot 2 - 1] = \left(\frac{1}{2}\right)^n \left[2^{-n} \cdot 2^n \cdot 2 - \left(\frac{1}{2}\right)^n \right]$$

$$h(n) = \left(\frac{1}{2}\right)^n \left[2 - \left(\frac{1}{2}\right)^n \right], \quad n \geq 0$$

Example 4.2.1

Determine the spectra of the signals

(a) $x(n) = \cos \sqrt{2}\pi n$

(b) $x(n) = \cos \pi n/3$

Solution

(a) For $\omega_0 = \sqrt{2}\pi$, we have $f_0 = 1/\sqrt{2}$. Since f_0 is not a rational number, the signal is not periodic. Consequently, this signal cannot be expanded in a Fourier series. Nevertheless, the signal does possess a spectrum. Its spectral content consists of the single frequency component at $\omega = \omega_0 = \sqrt{2}\pi$.

(b) In this case $f_0 = \frac{1}{6}$ and hence $x(n)$ is periodic with fundamental period $N = 6$. From (4.2.8) we have

$$c_k = \frac{1}{6} \sum_{n=0}^5 x(n) e^{-j2\pi kn/6} \quad k = 0, 1, \dots, 5$$

However, $x(n)$ can be expressed as

$$x(n) = \cos \frac{2\pi n}{6} = \frac{1}{2} e^{j2\pi n/6} + \frac{1}{2} e^{-j2\pi n/6}$$

which is already in the form of the exponential Fourier series in (4.2.7). In comparing the two exponential terms in $x(n)$ with (4.2.7), it is apparent that $c_1 = \frac{1}{2}$. The second exponential in $x(n)$ corresponds to the term $k = -1$ in (4.2.7). However, this term can also be written as

$$e^{-j2\pi n/6} = e^{j2\pi(5-n)/6} = e^{j2\pi(5n)/6}$$

which means that $c_{-1} = c_5$. But this is consistent with (4.2.9), and our previous observation that the Fourier series coefficients form a periodic sequence of

period N . Consequently, we conclude that

$$c_0 = c_2 = c_3 = c_4 = 0$$

$$c_1 = \frac{1}{2} \qquad c_5 = \frac{1}{2}$$

4.2.2 Power Density Spectrum of Periodic Signals

The average power of a discrete-time periodic signal with period N was defined in (2.1.23) as

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \quad (4.2.10)$$

We shall now derive an expression for P_x in terms of the Fourier coefficient $\{c_k\}$.

If we use the relation (4.2.7) in (4.2.10), we have

$$\begin{aligned} P_x &= \frac{1}{N} \sum_{n=0}^{N-1} x(n)x^*(n) \\ &= \frac{1}{N} \sum_{n=0}^{N-1} x(n) \left(\sum_{k=0}^{N-1} c_k^* e^{-j2\pi kn/N} \right) \end{aligned}$$

Now, we can interchange the order of the two summations and make use of (4.2.8), obtaining

$$\begin{aligned} P_x &= \sum_{k=0}^{N-1} c_k^* \left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right] \\ &= \sum_{k=0}^{N-1} |c_k|^2 = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 \end{aligned} \tag{4.2.11}$$

which is the desired expression for the average power in the periodic signal. In other words, the average power in the signal is the sum of the powers of the individual frequency components. We view (4.2.11) as a Parseval's relation for discrete-time periodic signals. The sequence $|c_k|^2$ for $k = 0, 1, \dots, N - 1$ is the distribution of power as a function of frequency and is called the *power density spectrum* of the periodic signal.

If we are interested in the energy of the sequence $x(n)$ over a single period, (4.2.11) implies that

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2 = N \sum_{k=0}^{N-1} |c_k|^2 \quad (4.2.12)$$

Example 4.2.2 Periodic “Square-Wave” Signal

Determine the Fourier series coefficients and the power density spectrum of the periodic signal shown in Fig. 4.11.

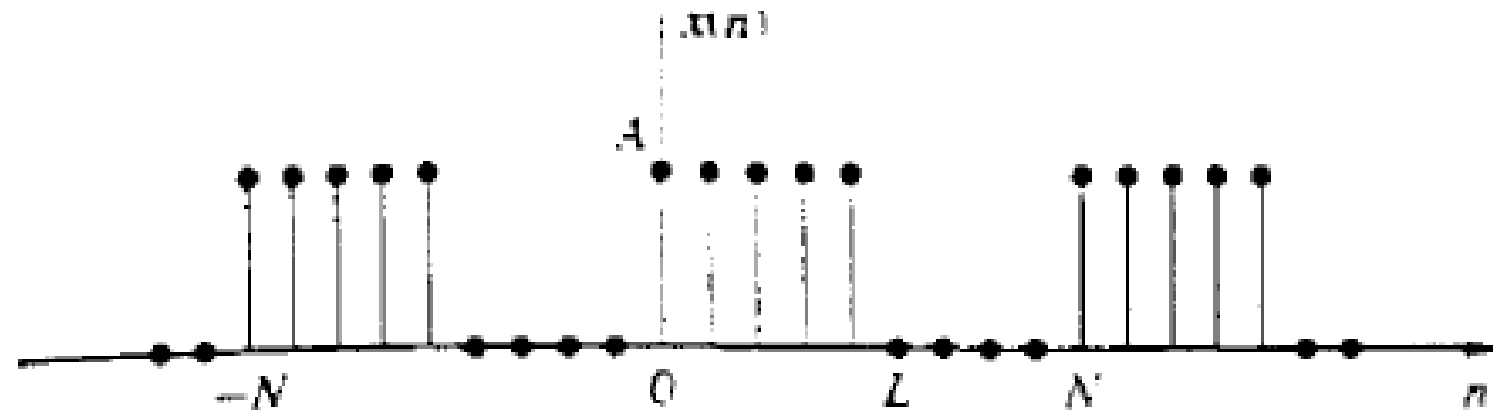


Figure 4.11 Discrete-time periodic square-wave signal.

Solution By applying the analysis equation (4.2.8) to the signal shown in Fig. 4.11, we obtain

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} = \frac{1}{N} \sum_{n=0}^{L-1} A e^{-j2\pi kn/N} \quad k = 0, 1, \dots, N-1$$

which is a geometric summation. Now we can use (4.2.3) to simplify the summation above. Thus we obtain

$$c_k = \frac{A}{N} \sum_{n=0}^{L-1} (e^{-j2\pi k/N})^n = \begin{cases} \frac{AL}{N}, & k = 0 \\ \frac{A}{N} \frac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}}, & k = 1, 2, \dots, N-1 \end{cases}$$

"partial summation formula" shown below:

$$\sum_{n=0}^{N-1} \alpha^n = \sum_{n=0}^{\infty} \alpha^n - \sum_{n=N}^{\infty} \alpha^n = \frac{1}{1-\alpha} - \frac{\alpha^N}{1-\alpha} = \frac{1-\alpha^N}{1-\alpha}$$

The last expression can be simplified further if we note that

$$\begin{aligned}\frac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}} &= \frac{e^{-j\pi kL/N} e^{j\pi kL/N} - e^{-j\pi kL/N}}{e^{-j\pi k/N} e^{j\pi k/N} - e^{-j\pi k/N}} \\ &= e^{-j\pi k(L-1)/N} \frac{\sin(\pi kL/N)}{\sin(\pi k/N)}\end{aligned}$$

Therefore,

$$c_k = \begin{cases} \frac{AL}{N}, & k = 0, +N, \pm 2N, \dots \\ \frac{A}{N} e^{-j\pi k(L-1)/N} \frac{\sin(\pi kL/N)}{\sin(\pi k/N)}, & \text{otherwise} \end{cases} \quad (4.2.21)$$

The power density spectrum of this periodic signal is

$$|c_k|^2 = \begin{cases} \left(\frac{AL}{N}\right)^2, & k = 0, +N, \pm 2N, \dots \\ \left(\frac{A}{N}\right)^2 \left(\frac{\sin \pi kL/N}{\sin \pi k/N}\right)^2, & \text{otherwise} \end{cases} \quad (4.2.22)$$

Figure 4.2.3 illustrates the plots of $|c_k|^2$ for $L = 2$, $N = 10$ and 40 , and $A = 1$.

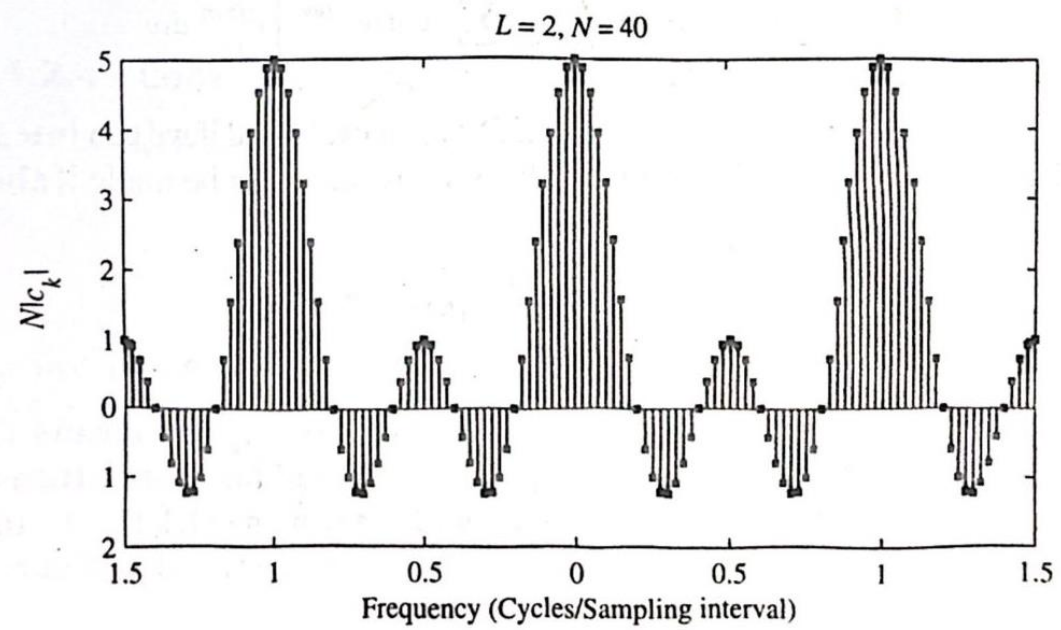
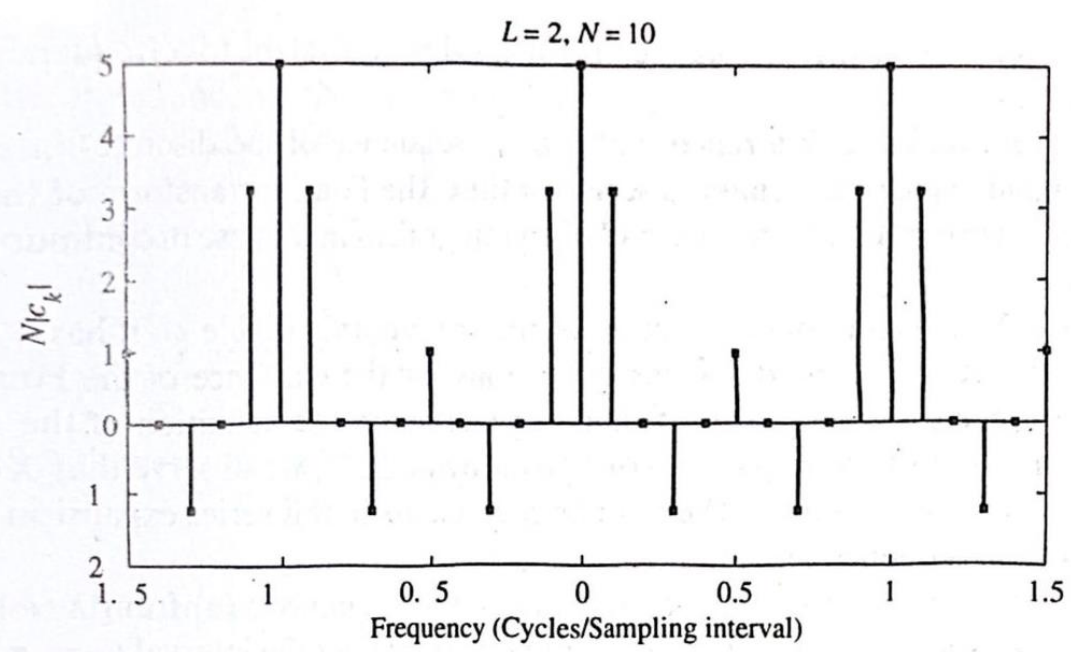


Figure 4.2.3 Plot of the power density spectrum given by (4.2.22).

4.2.3 The Fourier Transform of Discrete-Time Aperiodic Signals

Just as in the case of continuous-time aperiodic energy signals, the frequency analysis of discrete-time aperiodic finite-energy signals involves a Fourier transform of the time-domain signal. Consequently, the development in this section parallels to a large extent, that given in Section 4.1.3.

The Fourier transform of a finite-energy discrete-time signal $x(n)$ is defined as

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \quad (4.2.23)$$

Physically, $X(\omega)$ represents the frequency content of the signal $x(n)$. In other words, $X(\omega)$ is a decomposition of $x(n)$ into its frequency components.

Indeed, $X(\omega)$ is periodic with period 2π , that is,

$$\begin{aligned} X(\omega + 2\pi k) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j(\omega + 2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}e^{-j2\pi kn} \\ &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = X(\omega) \end{aligned} \quad (4.2.24)$$

Hence $X(\omega)$ is periodic with period 2π .

In summary, the *Fourier transform pair for discrete-time signals* is as follows.

FREQUENCY ANALYSIS OF DISCRETE-TIME APERIODIC SIGNALS

Synthesis equation
inverse transform

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega \quad (4.2.28)$$

Analysis equation
direct transform

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \quad (4.2.29)$$

4.2.5 Energy Density Spectrum of Aperiodic Signals

Recall that the energy of a discrete-time signal $x(n)$ is defined as

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad (4.2.40)$$

Let us now express the energy E_x in terms of the spectral characteristic $X(\omega)$. First we have

$$E_x = \sum_{n=-\infty}^{\infty} x(n)x^*(n) = \sum_{n=-\infty}^{\infty} x(n) \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) e^{-j\omega n} d\omega \right]$$

If we interchange the order of integration and summation in the equation above, we obtain

$$\begin{aligned} E_x &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X^*(\omega) \left[\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right] d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \end{aligned}$$

Therefore, the energy relation between $x(n)$ and $X(\omega)$ is

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(\omega)|^2 d\omega \quad (4.2.41)$$

This is Parseval's relation for discrete-time aperiodic signals with finite energy.

The spectrum $X(\omega)$ is, in general, a complex-valued function of frequency. It may be expressed as

$$X(\omega) = |X(\omega)|e^{j\Theta(\omega)} \quad (4.2.42)$$

where

$$\Theta(\omega) = \angle X(\omega)$$

is the phase spectrum and $|X(\omega)|$ is the magnitude spectrum.

As in the case of continuous-time signals, the quantity

$$S_{xx}(\omega) = |X(\omega)|^2 \quad (4.2.43)$$

represents the distribution of energy as a function of frequency, and it is called the *energy density spectrum* of $x(n)$. Clearly, $S_{xx}(\omega)$ does not contain any phase information.

Example 4.2.3

Determine and sketch the energy density spectrum $S_{xx}(\omega)$ of the signal

$$x(n) = a^n u(n) \quad -1 < a < 1$$

Solution Since $|a| < 1$, the sequence $x(n)$ is absolutely summable, as can be verified by applying the geometric summation formula,

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{\infty} |a|^n = \frac{1}{1 - |a|} < \infty$$

Hence the Fourier transform of $x(n)$ exists and is obtained by applying (4.2.29). Thus

$$X(\omega) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (ae^{-j\omega})^n$$

Since $|ae^{-j\omega}| = |a| < 1$, use of the geometric summation formula again yields

$$X(\omega) = \frac{1}{1 - ae^{-j\omega}}$$

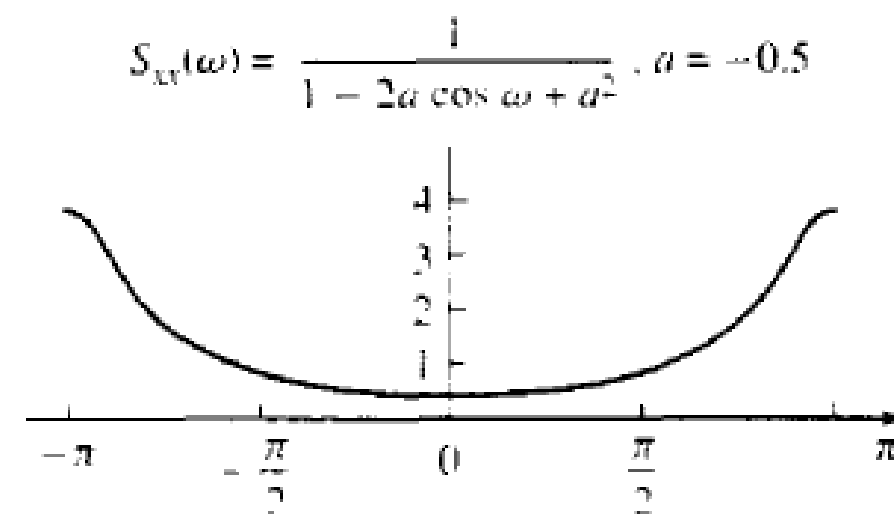
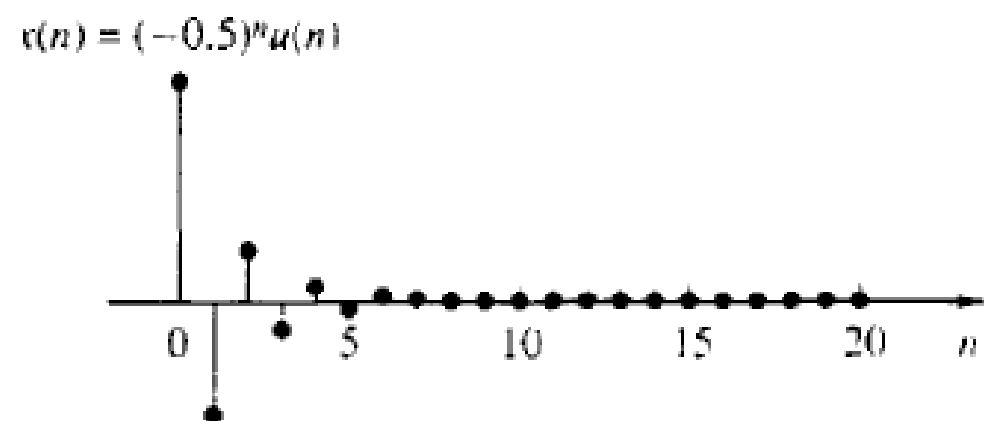
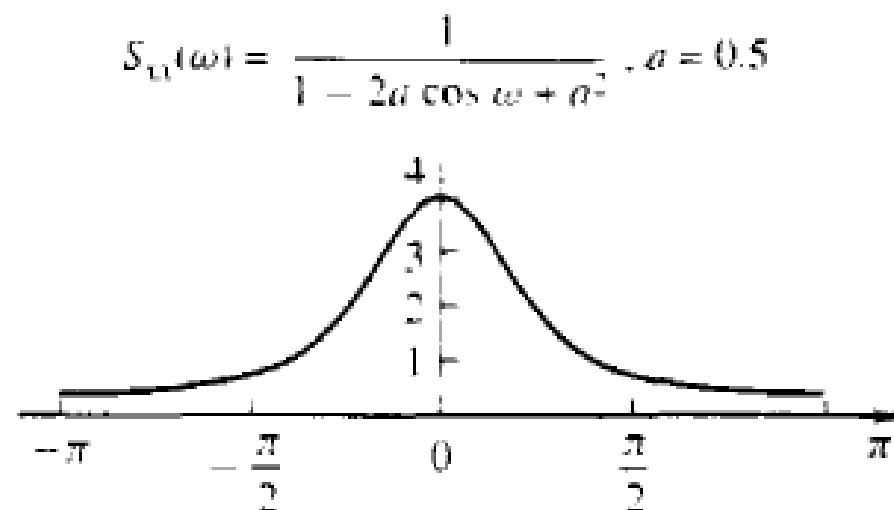
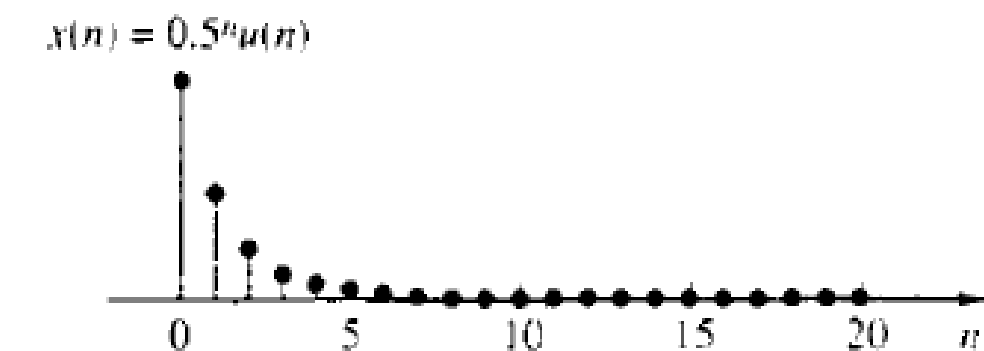
The energy density spectrum is given by

$$S_{xx}(\omega) = |X(\omega)|^2 = X(\omega)X^*(\omega) = \frac{1}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}$$

or, equivalently, as

$$S_{xx}(\omega) = \frac{1}{1 - 2a \cos \omega + a^2}$$

Figure 4.15 shows the signal $x(n)$ and its corresponding spectrum for $a = 0.5$ and $a = -0.5$.



(a)

(b)

Figure 4.15 (a) Sequence $x(n) = (\frac{1}{2})^n u(n)$ and $x(n) = (-\frac{1}{2})^n u(n)$; (b) their energy density spectra.